21-242 Recitation 12

Dimension — Q4. Extension lemma

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Extension Lemma: If $\dim(V) < \infty$, any linearly independent list can be extended to a basis.

Course-level solution

Let $(\vec{v}_1, \ldots, \vec{v}_k)$ be independent and let $(\vec{w}_1, \ldots, \vec{w}_n)$ be a basis, $n = \dim V$. Consider the concatenated list $(\vec{v}_1, \ldots, \vec{v}_k, \vec{w}_1, \ldots, \vec{w}_n)$. It spans V. Apply the extraction procedure of Span Q5: keep a vector iff it is not in the span of those already kept. All of $\vec{v}_1, \ldots, \vec{v}_k$ are kept (they start independent). The surviving list is independent, still spans V, and is therefore a basis extending the original list.

Later-course solution (using ideas from the rest of the handout)

If you already know that independent lists have length $\leq n$ and that a spanning list of length n is a basis, you may add one vector at a time from outside the current span; the process stops by length at latest at n, and must stop with a spanning list.